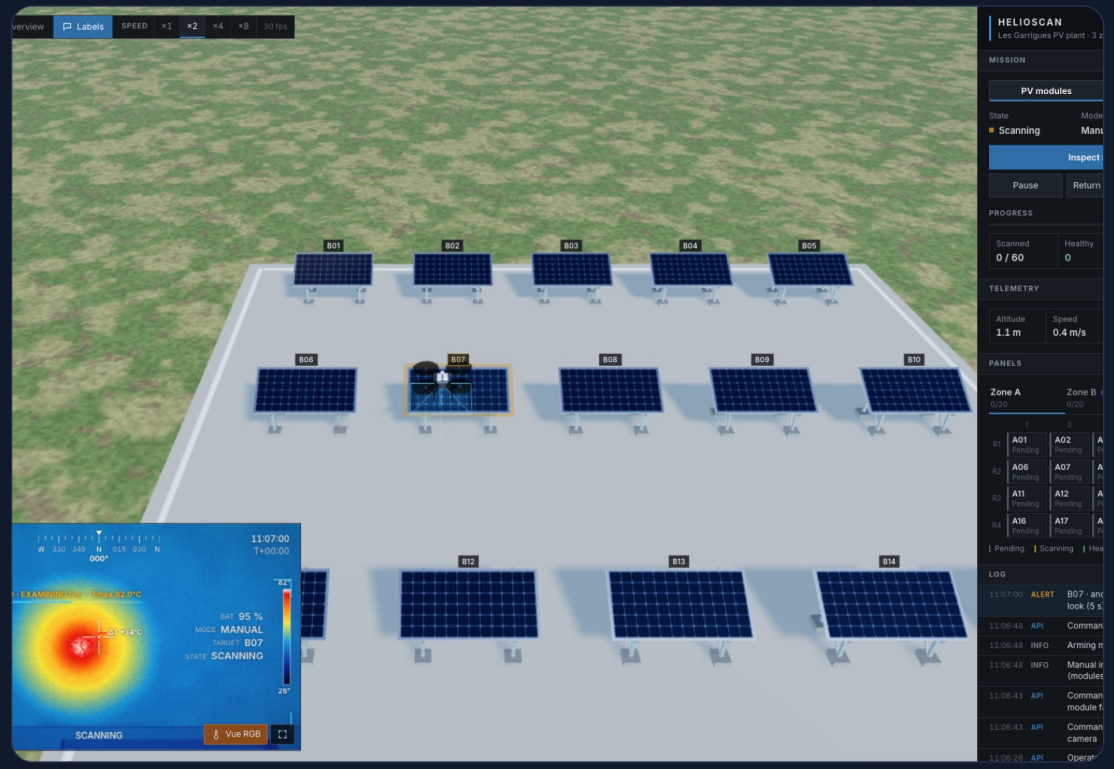
HACKTURIN 2026 · TORINO

HelioScan

Close-range drone inspection for solar plants, and the console that turns a fault into a work order

RGB + thermal · on-board processing · automatic work orders

3rd place · Beyond Solar track



CONTEXT

Costly defects, invisible from the ground

Hot spot

A shaded or damaged cell turns power into heat: lost output, faster ageing, fire risk

Cracked cell

Mechanical or thermal stress breaks the cell; inactive areas grow and often become hot spots

Soiling and bird droppings

Local shading creates a mismatch in the string and can itself trigger a hot spot

HOW IT'S CHECKED TODAY

Manual walk-throughs

Technicians with a handheld thermal camera: slow, tiring, easy to miss

Fixed camera towers

Usually rented; they watch the site, not each cell

High-altitude drone passes

The big picture, but one faulty cell is only a few pixels

APPROACH

Build the payload, not the drone

A stabilised RGB + thermal camera on a proven, off-the-shelf drone, flown close enough to read every panel the way a thermographer would

0.65 m

from the glass during the scan, not tens of metres above the plant

−60°

gimbal pitch: the camera looks perpendicular to a panel tilted at 30°

5 s

hold in front of each defect that comes into view, for a clean close-up

SYSTEM

One loop: detect, dispatch, inspect, act

1 · DETECT

Current drop

Module current falls below what the irradiance predicts

→

2 · ALERT

Alert opens

Watch, then critical if the drop deepens

→

3 · DISPATCH

Drone takes off

On its own; critical alerts go first

→

4 · INSPECT

Close-range scan

RGB + thermal, photo of the anomaly

→

5 · ACT

Work order

Action, priority, estimated loss

PLANT SIDE

Module currents, irradiance, wind (simulated in the demo)

ON THE DRONE

Stabilised RGB + thermal camera, close-range flight

OPERATOR CONSOLE

Supervision, alerts, auto-dispatch, work orders

PLATFORM

Three parts, one server in the middle

THE SITE

3D simulation

Plant, drone, mission logic and onboard camera

WebSocket

drone state 5 Hz, commands

THE BRAIN

Node.js server

Electrical supervision, alerts, dispatch, work orders

WebSocket

telemetry, alerts, live video

THE OPERATOR

Operator console

Supervision, flight control, alerts, work orders

Open by design

Every action in the interfaces is also a REST call, so existing maintenance software can drive HelioScan

Ready for a real drone

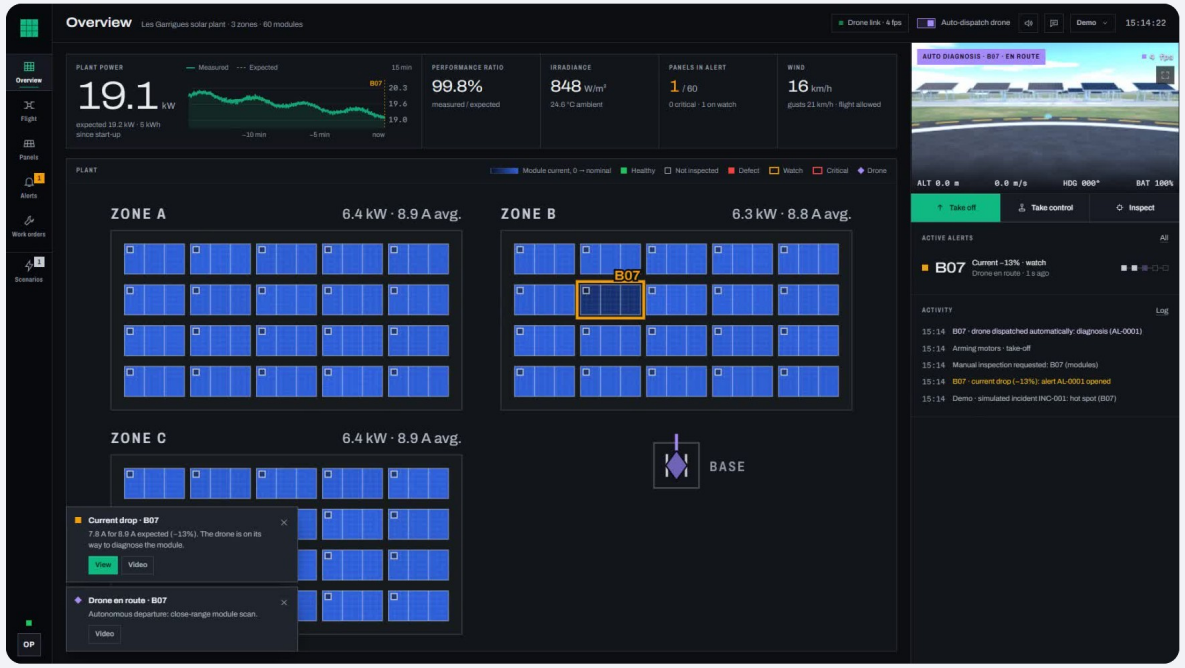
The console only talks to the server: the same console could supervise a real drone

PLATFORM

The current decides when the drone flies

Each module's current is compared with the current expected for its zone's irradiance. A passing cloud lowers both, so it triggers nothing.

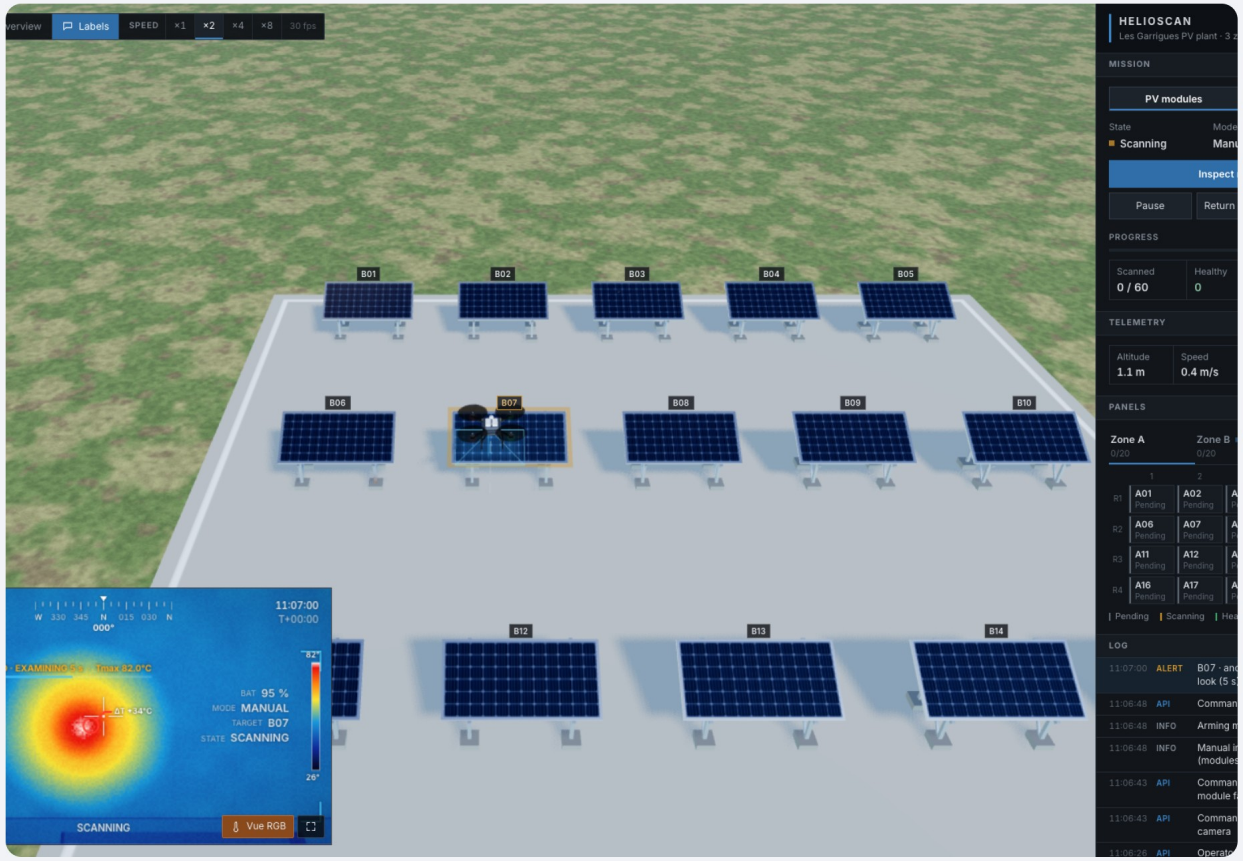
Deviation from expected	For	Effect
≤ −6 %	4 s	Watch alert
≤ −15 %	3 s	Escalates to critical
> −3 %	15 s	Alert closed



Overview: the module map darkens with measured current; B07 is on watch

SIMULATION

A close-range scan, thermal on board



Descent to 0.65 m

Gimbal at −60°, perpendicular to the 30° panel

U-shaped pass

Along both long edges of every module

Hold on the defect

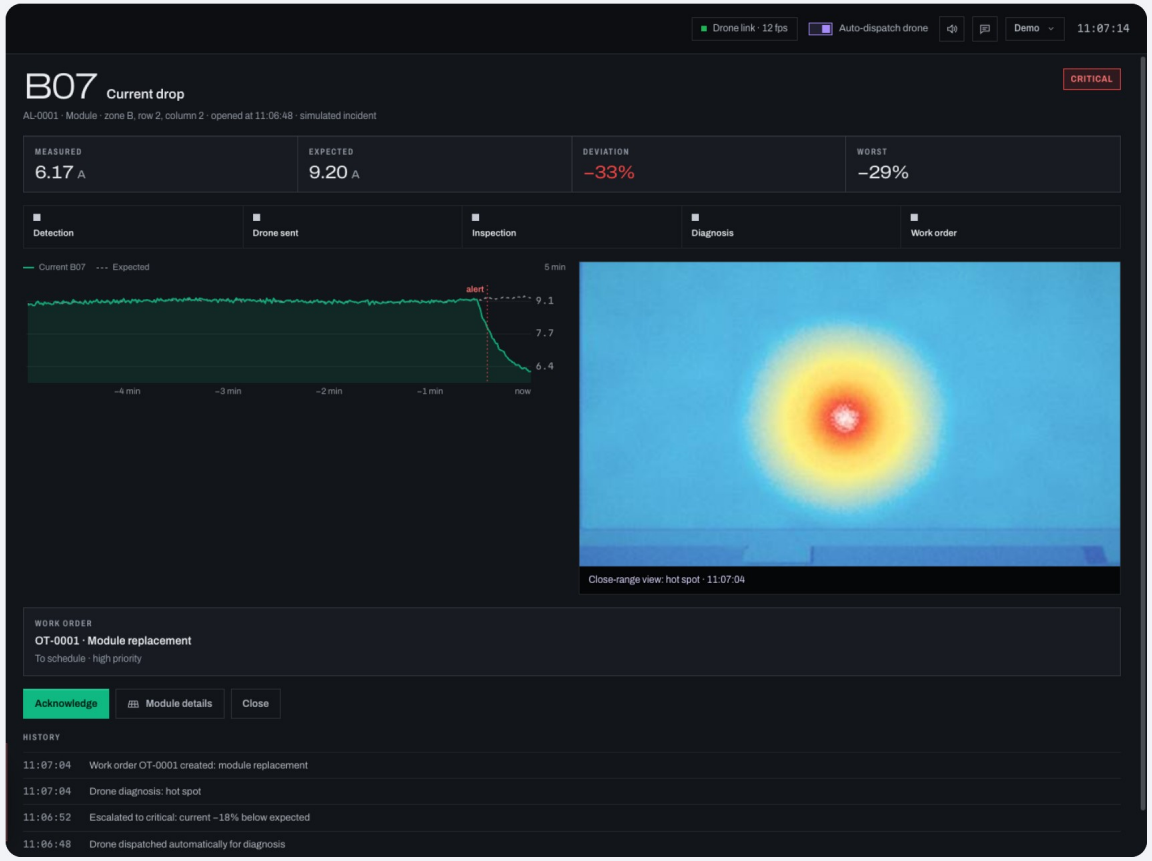
5 s in front of it, then the photo goes to the console

Structures too

Low flight in the aisles, a stop at each leg and foundation

SIMULATION · HOT SPOT ON B07

From a current drop to a work order



11:06:48

Current drop on B07: alert opens, drone dispatched automatically

11:06:52

Escalated to critical: current 18 % below expected

11:07:04

Drone diagnosis: hot spot, with its thermal close-up

11:07:04

Work order OT-0001: module replacement, high priority, about 2 h, 105 W lost

Demo run, simulation at ×2 speed, no operator input

SIMULATION

A full zone, inspected panel by panel



20 / 20
panels of zone B analysed

16 healthy · 4 defects
B03 crack, B07 hot spot, B11 and B20 soiling, each with its position

5:02
mission time for zone B (simulated clock)

SIMULATION

Twelve incidents, detected like on site

Family	Scenario	Detected by	What the drone does
Module	Hot spot, cracked cell, soiling, electrical fault	Module current (−12 % to −35 %)	Close-range scan
Structure	Abnormal tilt, foundation settling, corrosion	Tilt sensor, weather station	Low pass, stop at each leg
Structure	Wind gust over a zone	Anemometer	Waits for the gusts to end (no flight above 36 km/h), then inspects
Terrain	Heavy rain, vegetation shading, insulation fault, ground subsidence	Rain gauge, row currents, inverter, tilt sensors	Overflight with the camera pointing down

Each scenario places a real incident in the simulation; the console only sees what a sensor would report.

EDGE AI

Payload prototype: YOLO on the Arduino UNO Q

SENSOR HEAD

Phone camera

Stands in for the payload camera; shows the overlay

Wi-Fi · HTTPS

frames in, results out

EDGE COMPUTER · LINUX

Arduino UNO Q

- YOLO11n on ONNX Runtime
- Multi-object tracking
- U-path mission, reference check

Bridge RPC

defect events

MCU

STM32

LED alerts on the board

No cloud

All the vision runs on the board, as it would in the field

A printed test panel

6 A4 sheets with two bird droppings and a crack

The drone's path

Start on an ArUco marker, then top, right and bottom edges

EDGE AI

Synthetic training data, no hand labelling

Labels for free

Subtracting the healthy print from the defect print gives exact defect boxes

300 randomised phone views

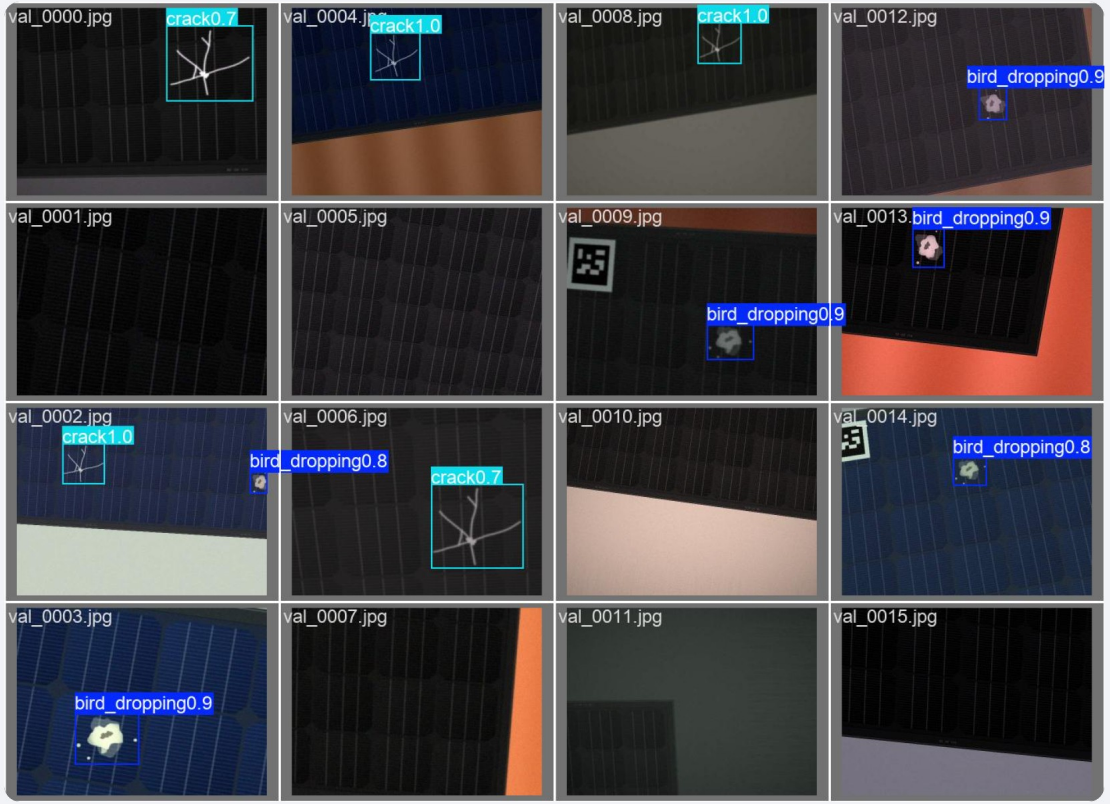
Distance, tilt, black-and-white or colour print, lighting, blur, noise

YOLO11n at 320 px

Two classes, fine-tuned from COCO in 15.8 min on a laptop CPU

One defect, one report

Tracking merges the many frames that see the same defect



Detector output on synthetic validation images

RESULTS

Every defect found, no false alarm

14 / 14

simulated inspection flights passed: 7 conditions × 2 prints

1.000

recall on the validation set

0.994

precision on the validation set

0.995

mAP50 (mAP50-95: 0.842)

Validation: 80 synthetic images never used in training. Flights: 3 defects found on the defect print and 0 on the healthy print, every time.